

19.	Answer: A Do a <u>scalar</u> sum of the EPE due to the interaction between the charge at the centre and the 5 charges surrounding it. $\text{EPE} = \frac{Q^2}{4\pi\epsilon_0 r}$ $\text{WD by external force} = \text{Change in EPE} = \text{EPE}_{\text{infinity}} - \text{EPE}_{\text{initial}} = 0 - \frac{Q^2}{4\pi\epsilon_0 r}$
20.	Answer: D